The documentation and process conversion measures necessary to comply with this revision shall be completed by 24 February 2010.

INCH-POUND

MIL-PRF-19500/554E
24 November 2009
SUPERSEDING
MIL-PRF-19500/554D
25 May 2006

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, SCHOTTKY BARRIER, FAST RECOVERY, TYPE 1N6392, JAN, JANTX, JANTXV, AND JANHC

This specification is approved for use by the all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for a silicon, fast recovery, Schottky barrier semiconductor diode. Three levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500. One level of product assurance is provided for each unencapsulated device type.

1.2 Physical dimensions. See figure 1 (DO-5) and figure 2 (die).

1.3 Maximum ratings. Unless otherwise specified T_A = 25°C.

Type	V_{RRM} and V_{RWM} (1)	V_{RSM}	V_R (1)	I_{FM} (2) T_C = +115°C	I_O (3) T_C = +115°C	I_{FSM}	T_J T_{STG}	Max $R_{\Theta JC}$
	V(pk)	V(pk)	V dc	A dc	A dc	A (pk)	°C	°C/W
1N6392	45	54	45	60	54	1,000	-55 to +175	1.0

(1) Full rated V_{RRM} and V_{RWM} is applicable over the range of T_C = -55°C to + 173°C for I_{FM} = 0. Full rated V_R is applicable over the temperature range of T_C = -55°C to +166°C. When V_R = 45 V dc and T_C = +166°C, then T_J = +175°C.

(2) Average current with a 50 percent duty cycle square wave including reverse amplitude equal to the magnitude of full rated V_{RWM}. Derate linearly at 1.09 A dc/°C for T_C > +115°C.

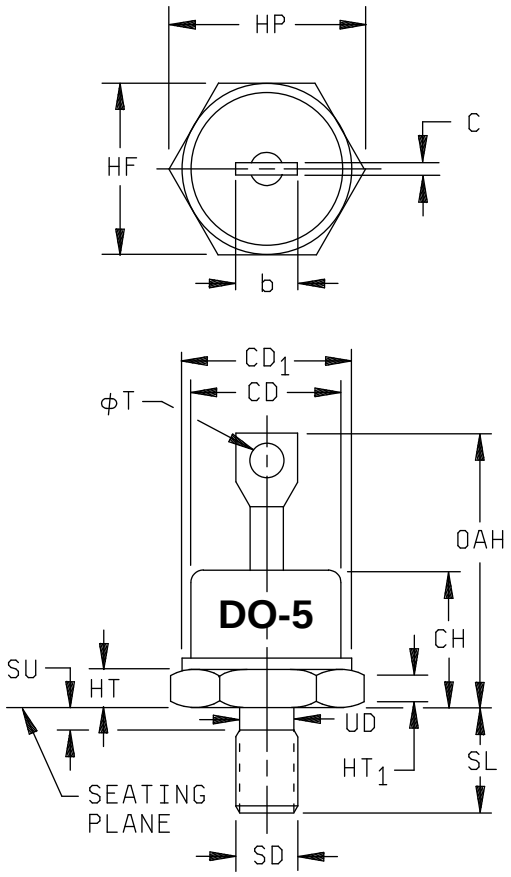
(3) Average current with an applied sine wave peak value equal to the magnitude of full rated V_{RWM}. For temperature-current derating curves, see figure 3.

1.4 Unless otherwise noted, primary electrical characteristics at T_C = +25°C.

Type	Max V_{FM1} I_{FM} = 120 A (pk)	Max V_{FM3} I_{FM} = 10 A (pk)	Max I_{RM} V_{RM} = 45 V (pk)		Max C_T V_R = 5 V dc
			T_J = +25°C	T_J = +175°C	
	V (pk)	V (pk)	mA (pk)	mA (pk)	pF
1N6392	0.82	0.51	2.0	200	3,000

* Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil .

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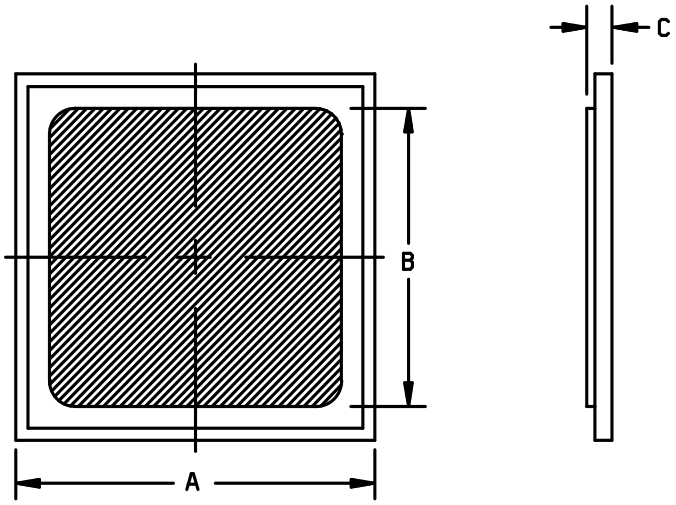


Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
C		.375		9.53	7
CD		.667		16.94	
CH		.450		11.43	
C1	.025	.080	0.64	2.03	
HF	.669	.688	17.00	17.48	
HT1	.115	.200	2.92	5.08	
HT2	.060		1.52		6
OAH	.750	1.00	19.05	25.40	
SD					5
SL	.422	.453	10.72	11.51	
SU		.090		2.29	4
UD	.220	.249	5.59	6.32	
ΦT	.140	.175	3.56	4.45	

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Units must not be damaged by torque of 30 inch-pound applied to .25-28 UNF-2B nut assembled on thread.
4. Length of incomplete or undercut threads of UD.
5. Maximum pitch diameter of plated threads shall be basic pitch diameter .2268 inch (5.76 mm) reference (FED-STD-H28, “Screw-Thread Standards for Federal Services”).
6. A chamfer or undercut on one or both ends of the hex portion is optional; minimum base diameter at seating plane .600 inch (15.24 mm).
7. The angular orientation and peripheral configuration of terminal 1 is undefined, however, the major surfaces over dimensions C and C_1 shall be flat.
8. The anode is connected to terminal 1. The cathode is connected to terminal 2.
9. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

FIGURE 1. Physical dimensions (DO-5).

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Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.210	.220	5.38	5.46
B	.190	.210	5.03	5.08
C	.012	.013	.30	.33

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. The metallization characteristics of the die are:
   Anode (front) Cathode (back)
   Ag Ag
4. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

* FIGURE 2. JANHCA die dimensions (A version).

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch or https://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

* 2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and figures 1 and 2 herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Diode construction. These devices shall be constructed in a manner and using material which enable the diodes to meet the applicable requirements of MIL-PRF-19500 and this document.

3.5.1 Polarity. Devices shall have the cathode electrically connected to the stud.

3.6 Marking. Marking shall be in accordance with MIL-PRF-19500.

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3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.8 Electrical test requirements. The electrical test requirements shall be specified in table I herein.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or requalification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.2.2 JANHC devices. JANHC devices shall be qualified in accordance with MIL-PRF-19500.

* 4.3 Screening (JANTXV and JANTX levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (table E-IV of MIL-PRF-19500)	Measurement
	JANTXV and JANTX levels
3a	Condition C
(1) 3b	Surge, see 4.3.2
3c	Thermal impedance, see 4.3.3
4	Not applicable
9	Not applicable
10	Not applicable
(2) 11	Reverse energy test (see 4.5.2), followed by V_{FM1} and I_{RM1} of table I, subgroup 2.
12	Burn-in, method 1038 of MIL-STD-750, test condition A; see 4.3.4.
(3) 13	F_{M1} and I_{RM1} of table 1, subgroup 2; ΔV_{FM1} = 0.05 V (pk), ΔI_{RM1} = ±100 percent or 5 mA dc, whichever is greater. Scope display evaluation (see 4.5.3).

(1) Surge current shall precede thermal response. These tests shall be performed anytime after screen 3a and before screen 10.
(2) I_{RM1} shall not be indicative of an open condition.
(3) All devices shall be subjected to the scope display evaluation test, and reverse energy test.

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4.3.1 Screening (JANHC). Screening of JANHC die shall be in accordance with MIL-PRF-19500.

4.3.2 Surge current. Surge current shall be performed in accordance with method 4066 of MIL-STD-750, with the following conditions: I_{FSM} = 1,000 A, 6 surges, t_p = 8.3 ms, 1/2 sine wave, or rectangular pulse of equivalent I_{RMS}, I_O = 0, V_{RWM} = 0, duty factor one percent minimum, T_A = 25°C.

* 4.3.3 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3101 or 4081 as applicable of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{SW} (V_C and V_H where appropriate). Measurement delay time (t_{MD}) = 300 μs max. See table II, subgroup 4 herein.

* 4.3.4 Burn-in. Method 1038 of MIL-STD-750, test condition A. T_J = 150°C minimum, V_R = 36V dc, I_O = 0, t = 48 hours.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

* 4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

Subgroup	Method	Condition
B3	1048	T_J = 150°C minimum, V_R = 36 V dc. I_O = 0, t = 48 hours.
B5	4081	See 4.3.3.
B7	4066	I_{FSM} = 1,000 A (pk); V_{RM} = 45 V (pk), I_O = 45 A dc; 10 surges of 8.3 ms each at 1 minute intervals; 100°C < T_C < 125°C; 22 devices, c = 0.

* 4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

Subgroup	Method	Condition
C2	2036	Test condition A, 15 lbs, t = 15 s. Test condition F (method B), 15 pounds, t = 15 s. Test condition D1, 3 inch-pound, t = 15 s. Test condition D2, 30 inch-pound, t = 15 s.
C5	4081	$R_{\Theta JC}$ = 1.0°C/W maximum in accordance with 4.3.3 except t_H = 25 s (minimum), n = 22, c = 0.
C6	1048	T_J = 150°C minimum, V_R = 36 V dc. I_O = 0.
C6	1037	Operational power cycling, I_F = 50 A dc, $T_{C\ (low)}$ = +40°C +0°C, -15°C; $T_{C\ (high)}$ = 115°C +5°C, -0°C; 6,000 cycles; 22 devices, c = 0, (see 4.5.5).

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with MIL-PRF-19500 and the conditions for subgroup testing in table II herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

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4.5 Methods of inspection. Methods of inspection shall be as specified in appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

4.5.2 Peak reverse energy test. The peak reverse energy test is to be performed as on figure 4. The Schottky rectifier under test must be capable of absorbing the reverse energy, as defined, and meet the electrical requirements of table I, subgroup 4 herein.

4.5.3 Scope display evaluation. A curve trace visual display of the reverse characteristics (voltage versus current) shall be made using a calibration of 10 mA per division vertical sensitivity, and 10 V dc per division horizontal sensitivity. Increase the reverse voltage until the reverse current reaches 50 mA minimum to 70 mA maximum. Observe the curve characteristics. Any device with an erratic pattern, such as double trace, other than capacitance effects, intermittence, discontinuities other than zener noise at the knee of the curve, or shorts shall be removed from the lot. Time limit for this test shall be 5 seconds minimum. This test shall be the last test performed on the device in the 100 percent screening tests. The percent defective result for evaluation against the percent defective allowed (PDA) shall be determined prior to the results of this test.

4.5.4 Reverse current at peak reverse voltage, alternate test. The reverse current at peak reverse voltage test requirement may be satisfied by performing the reverse energy test of 4.5.2 and measuring breakdown voltage to ensure $V_{BR} \geq 54$ V (pk) with $I_{RM} = 2.0$ A (pk).

* 4.5.5 DC intermittent operation life. Method 1037 of MIL-STD-750. One complete cycle for this test shall consist of the following two steps: Step 1, heat the case to the T_C (high) specified by passing the specified forward current through the diode under test. The reverse voltage shall be only enough to permit the reverse current to flow, and in any event should be a maximum of 5 V dc. Step 2, remove the applied current and allow the case temperature to cool to the T_C (low) specified. No time limit is applicable to the steps, but the cycling must be continuous until the required number of cycles has been completed. It is permissible to force cool the device during step 2.

This test shall be considered a qualification test only, and needs to be performed during the initial qualification and as applicable to verify the adequacy of a design change.

4.5.6 Inspection conditions. Unless otherwise specified in MIL-PRF-19500 or herein, all inspections shall be conducted at a case temperature (T_C) of +25°C ± 3°C.

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TABLE I. Group A inspection.

nspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance	3101	See 4.3.3				°C/W
Forward voltage	4011	I_{FM} = 120 A (pk) pulsed (see 4.5.1)	V_{FM1}		0.82	V (pk)
		I_{FM} = 60 A (pk) pulsed (see 4.5.1)	V_{FM2}		0.68	V (pk)
		I_{FM} = 10 A (pk) pulsed (see 4.5.1)	V_{FM3}		0.51	V (pk)
Reverse current	4016	V_{RM} = 45 V (pk) pulsed (see 4.5.1)	I_{RM1}		2.0	mA (pk)
Subgroup 3						
High temperature operation						
Reverse current leakage	4016	V_{RM} = 45 V (pk) pulsed (see 4.5.1)				
		T_J = +175°C	I_{RM2}		200	mA (pk)
		T_J = +125°C	I_{RM3}		60	mA (pk)
Low temperature operation		T_C = -55°C				
Reverse current leakage	4016	V_{RM} = 45 V (pk) pulsed (see 4.5.1)	I_{RM4}		400	mA (pk)
Forward voltage	4011	I_{FM} = 10 A (pk) pulsed (see 4.5.1)	V_{FM4}		0.69	V (pk)
Subgroup 4						
Reverse current leakage at peak reverse voltage	4016	Peak reverse energy test (see 4.5.2) V_{RSM} = 54 V (pk) (alternate test, see 4.5.4)	I_{RM5}		2.0	A (pk)
Capacitance	4001	V_R = 5 V dc; f = 100 KHz ≤ f ≤ 1 Mhz	C_T		3,000	pF

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5						
Not applicable						
Subgroup 6						
Forward surge	4066	I_{FSM} = 1,000 A (pk); 10 surges of 8.3 ms each at 1 minute intervals, superimposed on I_O = 22.5 A dc; V_{RWM} = rated V_{RWM} (see 1.3); T_A = +100°C.				
Electrical measurements		See table I, subgroup 2				
Subgroup 7						
Not applicable						

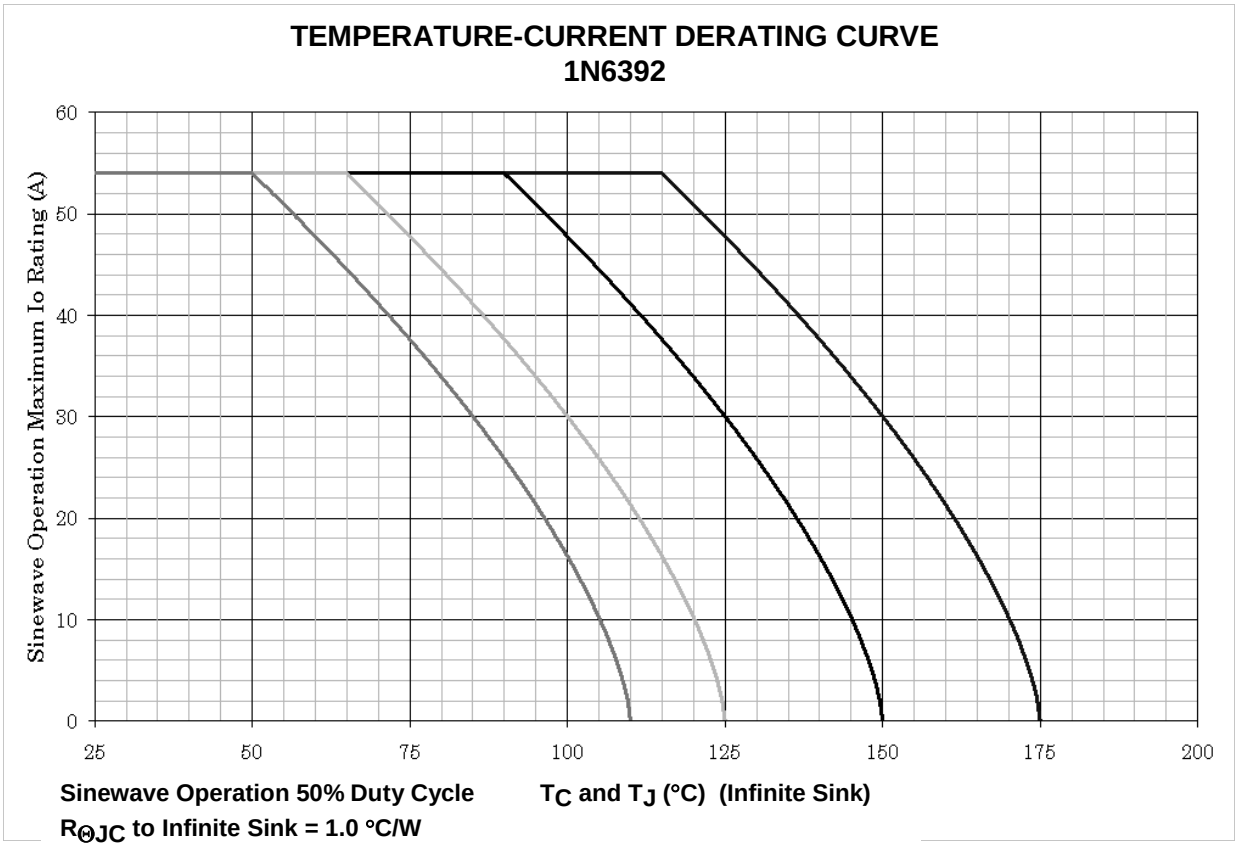
1/ For sampling plan, see MIL-PRF-19500.

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* TABLE II. Group E inspection (all quality levels) for qualification and requalification only.

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			22 devices c = 0
Thermal shock	1056	100 cycles, 0° to 100°C.	
Hermetic seal	1071	Test condition C and H_1.	
Electrical measurement		See table I, subgroup 2.	
Subgroup 2			22 devices c = 0
Steady-state reverse bias	1038	Test condition A, 1,000 hours.	
Electrical measurements		See table I, subgroup 2.	
Subgroup 4			Sample size = N/A
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 5			
Not applicable			
Subgroup 10			
Surge	4066	Condition A; T_A = +25°C; I_{FSM} = 1000 A, 100 surges	
Electrical measurements		See table I, subgroup 2.	

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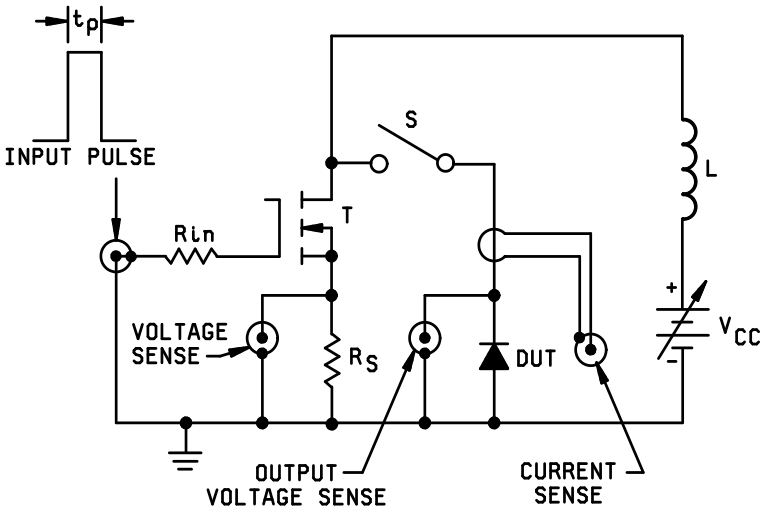
NOTE: Max finish-alloy temp = 175°C

NOTES:

1. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 175°C$) and current rating specified. (See 1.3 herein.)
3. Derate design curves chosen at $T_J \leq$, 150°C, 125°C, and 110°C to show current rating where most users want to limit T_J in their application.

* FIGURE 3. Temperature current derating curve.

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Input pulse	Rin = 50 ohms, 1 W
V_G = 10 V dc	R_S = 0.1 ohms, 1 W
R_G = 50 ohms	V_{CC} ≈ 10 V dc
P.W. ≈ 30 μs	L = 260 μH
Duty cycle ≤ 1 percent	T = IRF130/2N6756 or equivalent

PROCEDURES:

1. With S open, adjust pulse width to test current of 2 A across R_S.
2. Close S, verify test current with current sense.
3. Read peak output voltage (see 4.5.4).

FIGURE 4. Peak reverse energy test circuit.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

* 6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.daps.dla.mil .

* 6.4 Suppliers of JANHC die. The qualified JANHC suppliers with the applicable letter version (example, JANHCA1N6392) will be identified on the QML.

JANHC ordering information	
PIN	Manufacturer
	43611
1N6392	JANHCA1N6392

6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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